Sentiment Analysis of Movie Reviews Using Machine Learning

1. Introduction

This project conducts sentiment analysis on the IMDB movie review dataset, which contains 50,000 reviews labeled as positive or negative. The dataset is widely used in natural language processing research and provides a balanced collection of user-generated content. Sentiment analysis of movie reviews has practical applications in recommendation systems, box office prediction, and content moderation. The primary objective of this study is to implement and compare different machine learning algorithms for binary sentiment classification, with the goal of accurately predicting whether a review expresses positive or negative sentiment.

2. Literature Review

Sentiment analysis has evolved significantly through advancements in natural language processing and machine learning. Early approaches to sentiment classification relied on lexicon-based methods and simple statistical models. With the emergence of machine learning, algorithms such as Naive Bayes, Support Vector Machines (SVM), and ensemble methods have become standard for text classification tasks. Research by Pang and Lee (2004) demonstrated the effectiveness of machine learning techniques for movie review sentiment analysis, establishing benchmarks for subsequent studies. More recent approaches incorporate deep learning models and transformer architectures, but traditional machine learning methods remain relevant due to their interpretability and computational efficiency, particularly for moderate-sized datasets.

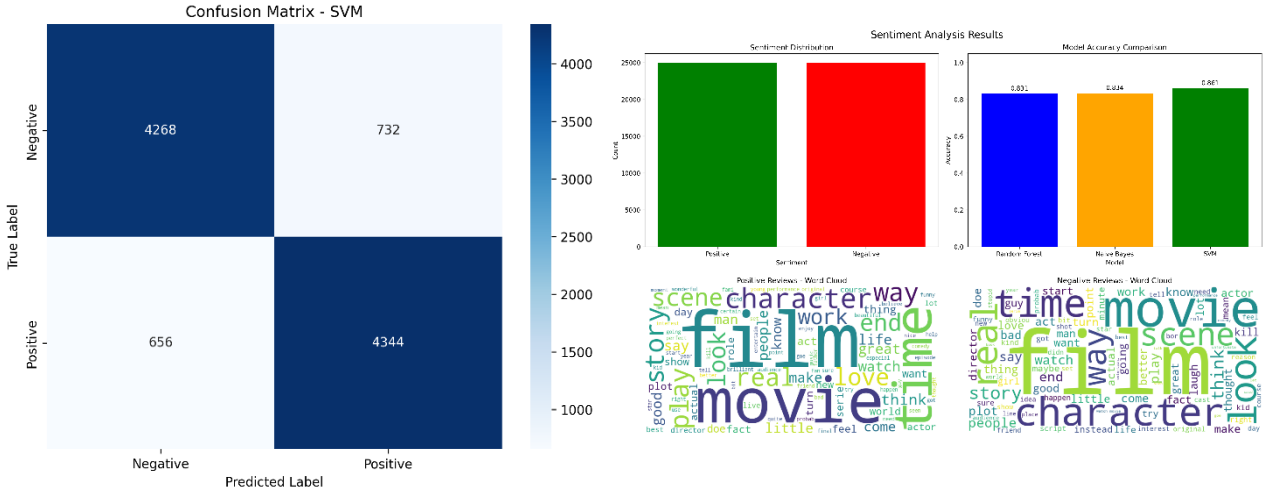
3. Methodology and Analysis

Text Preprocessing: The analysis began with comprehensive text cleaning including lowercase conversion, punctuation removal, and elimination of numerical characters. A custom tokenization approach was implemented without external NLP libraries. Stopwords were removed using scikit-learn's English stopword list, and a simple stemming algorithm was applied to reduce words to their root forms. The text was then vectorized using TF-IDF with unigram and bigram features, producing a feature space of 1,000 dimensions.

Modeling: Three machine learning models were implemented and compared: Random Forest, Naive Bayes, and Support Vector Machine (SVM). The Random Forest classifier used 100 trees with parallel processing enabled. The SVM employed a linear kernel, which is suitable for high-dimensional text data. All models were trained on 80% of the data (40,000 reviews) and evaluated on the remaining 20% (10,000 reviews).

Analysis Results: The models achieved the following performance metrics: Random

Forest(83.13% accuracy), Naive Bayes(83.44% accuracy), and SVM(86.12% accuracy).(presented on the right) The SVM demonstrated the best overall performance with 85.58% precision and 86.88% recall for positive sentiment classification. Analysis of the confusion matrix revealed that the SVM model correctly classified 4,344 positive reviews and 4,268 negative reviews out of 10,000 test samples.(presented on the left) Visualization of word frequency distributions showed distinct linguistic patterns between positive and negative reviews, with positive reviews containing terms like "excellent," "wonderful," and "best," while negative reviews featured words such as "terrible," "boring," and "disappointing."



4. Research Proposal

Future research could explore several enhancements to improve model performance and applicability. First, transformer-based models such as BERT or RoBERTa could be implemented to capture deeper contextual relationships in review text. These models have demonstrated state-of-the-art performance on various NLP tasks and could potentially improve accuracy beyond 90%. Second, incorporating additional features such as review metadata, user ratings, or temporal information could provide supplementary signals for sentiment classification. Third, expanding the classification framework to include neutral sentiment or fine-grained rating prediction (1-5 stars) would better reflect real-world review systems. Finally, developing a real-time sentiment analysis API would demonstrate practical application in content moderation and market research scenarios.

Data Source: The IMDB movie review dataset is publicly available and was obtained from Kaggle. This dataset contains 50,000 polar movie reviews for binary sentiment classification.